Spatially Aware Label-Error Detection for Geospatial Machine-Learning Training Data

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Abstract

Training labels for land-cover and other geospatial models are often wrong, and the errors are rarely independent. One annotator or one misleading image can corrupt every label in an area. We study how standard label-error detectors behave under such spatially clustered errors, using a synthetic land-cover benchmark and two real hyperspectral scenes (Indian Pines and Pavia University) with injected uniform, class-pair and clustered noise. Three findings hold on all datasets. First, the common neighbour-vote check, near-perfect for isolated errors on the real scenes, is close to chance on clustered errors (average precision 0.22–0.28 at a 20% error rate, where chance is 0.20). Second, a simple spatially smoothed variant of confident learning's self-confidence score, which averages out-of-fold class probabilities over each sample's ten nearest neighbours, improves average precision in 180 of 180 synthetic and 107 of 120 real comparisons (Wilcoxon $p < 0.05$ in every noise regime). Third, clustered errors remain hard: the best real-scene average precision is 0.31–0.41, and removing the flagged samples lowers downstream accuracy. A fourth finding does *not* transfer. On synthetic data, spatial block cross-validation recovers clustered errors that random cross-validation hides (15 of 15 runs). On the real scenes it is worse in every regime, because held-out blocks there are hard to predict even with clean labels: accuracy drops by 24 points against 1 point on the synthetic landscape. We propose this random-versus-spatial accuracy gap as a diagnostic for choosing the fold design. The methods are released as `geolabelqa`, an open-source Python package that also validates annotation polygon geometry.

1 Introduction

Supervised land-cover, crop-type and building-extraction models are trained on labels drawn by people: field surveyors, photo-interpreters and, increasingly, crowds of annotators working from imagery. Those labels contain errors, and the cost of the errors is well documented. Label noise lowers the accuracy of the resulting maps [10, 14], and noisy *evaluation* labels can reorder which model looks best [11]. Finding and fixing wrong labels before training is therefore a routine part of dataset curation.

The standard tools for finding them come from general machine learning. Classifier filtering [2] and confident learning [12] train a model on part of the data, predict the rest, and flag samples whose given label the model finds unlikely. A second family of checks is specific to spatial data and older than either. It appeals to Tobler's first law [21]: a sample whose label disagrees with most of its geographic neighbours is suspicious.

Both families assume that errors are *independent*. In geospatial annotation they often are not. One annotator may consistently confuse cropland with grassland across their assigned tiles. A

cloudy or out-of-date image may mislead every interpreter who labels a given area. A legacy map used as reference may be wrong for a whole district. Errors of this kind arrive in **spatial clusters**, and both families of checks are blind to them for the same reason. The neighbourhood check fails because the neighbours share the error. The model-based check fails whenever the model can learn the local error from nearby training samples. That happens when coordinates are features, which is common practice, or when features are themselves spatially autocorrelated. Out-of-fold predictions from *random* cross-validation then reproduce the mistake instead of contradicting it.

This is the label-auditing counterpart of a problem the mapping community knows well from model *evaluation*. Random cross-validation on spatially autocorrelated data gives over-optimistic accuracy estimates, and spatial block cross-validation is the accepted remedy [9, 15, 17, 23]. We test whether that remedy carries over to label auditing. It does on a synthetic landscape, but not on two real scenes. The reason is instructive: on real data, spatial blocks hide whole fields whose spectra the model cannot predict from elsewhere, so the auditor loses more than the leakage it avoids. One further inexpensive step helps consistently on all data: smoothing out-of-fold predictions over geographic neighbours.

Contributions. To our knowledge, this is the first evaluation of label-error detectors for geolocated point samples under spatially clustered annotator noise, with spatially blocked out-of-fold predictions. Specifically:

1. A controlled benchmark that separates three noise regimes (uniform, class-pair and spatially clustered) on a synthetic land-cover task with known ground truth, and the same regimes injected into two real hyperspectral scenes with expert ground truth (Section 4).

2. An empirical analysis, with paired significance tests, of when neighbour voting and confident learning fail, and of when spatial block CV helps or hurts. We trace the difference to a measurable random-versus-spatial accuracy gap (Sections 5 and 6).

3. *Smoothed self-confidence*: a simple, model-agnostic, spatially smoothed variant of the self-confidence score. It averages out-of-fold class probabilities over a sample's spatial neighbours before ranking (Section 3).

4. `geolabelqa`, an open-source Python package that implements the pipeline, together with a dependency-light validator for annotation polygons.

2 Related work

Label noise. Frénay and Verleysen [5] survey learning under label noise and distinguish noise that is completely at random, noise that depends on the class, and noise that depends on the features. Brodley and Friedl [2] introduced classifier filtering, which drops training samples that an ensemble trained on other folds misclassifies. Confident learning [12] estimates the joint distribution of given and true labels from out-of-fold probabilities with per-class thresholds. It flags off-diagonal samples and ranks them by self-confidence, the predicted probability of the given label. These methods treat samples as exchangeable and say nothing about how to build folds when samples are spatially dependent.

Label noise in remote sensing. Mnih and Hinton [10] trained road and building detectors from misregistered and incomplete map labels with noise-aware losses. Pelletier et al. [14] measured how random and systematic label noise degrade land-cover classification from satellite image time

series. Random forests tolerated substantial random noise, and systematic noise, where one class is consistently relabelled as another, was more damaging. Mellor et al. [8] used the random-forest ensemble margin to find mislabelled land-cover samples, and Santos et al. [18] cleaned satellite time-series samples with self-organising maps and Bayesian inference in feature space. The work closest to ours is Kröber et al. [7]. They benchmark data-centric label-noise identification methods, including confident learning, on remote sensing datasets, but do not consider spatially structured noise or the construction of the folds. Crowdsourced land-cover labels vary in quality between contributors [19], which is one source of the regionally systematic errors we model. Our class-pair and clustered regimes extend the systematic setting of Pelletier et al. [14] by adding spatial structure.

Spatial context in label cleaning. For hyperspectral images, several methods use spatial context to detect or correct noisy labels. Jiang et al. [6] propagate labels over a superpixel-based spectral–spatial graph, and Tu et al. [22] weight distances between superpixels and pixels. These methods operate on dense image grids and assume that pixels in the same local region share a class. That assumption is reasonable for isolated errors but breaks down when the errors themselves fill a region, which is the regime we study. For semantic segmentation, Tadokoro et al. [20] model spatially correlated label noise with a latent Gaussian field. Smoothed self-confidence belongs to this spatial-smoothing lineage. It differs in operating on sparse point samples and in smoothing *out-of-fold* predictions from spatially blocked models rather than labels or in-sample features, which is what keeps it informative when errors are clustered.

Spatial cross-validation. Spatial autocorrelation violates the independence that random K-fold assumes. Roberts et al. [17] recommend blocking for structured data, and `blockCV` [23] made spatial blocking routine in species-distribution modelling. Ploton et al. [15] showed that random-CV accuracy for large-scale biomass maps collapsed under spatial validation. Meyer et al. [9] found that coordinates and other location-proxy predictors encourage models to reproduce training data instead of generalising. These works concern accuracy assessment. We found no prior use of spatially blocked folds to produce the out-of-fold predictions on which label-error detection relies, and carry the observations over to that setting.

3 Method

Let $\{(\mathbf{x}_i, \mathbf{s}_i, \tilde{y}_i)\}_{i=1}^{n}$ be training samples with features $\mathbf{x}_i$, location $\mathbf{s}_i \in \mathbb{R}^2$ in a projected CRS, and given (possibly wrong) label $\tilde{y}_i \in \{1, \ldots, K\}$.

Out-of-fold probabilities. We partition the samples into F folds, train a classifier on $F-1$ folds and predict class probabilities $\hat{\mathbf{p}}_i \in \Delta^{K-1}$ for the held-out fold. With *random* CV, folds are a uniform random partition. With *spatial block* CV, the study area is tiled into square blocks of side b, and whole blocks are assigned to folds. Assignment is greedy, largest block first, which balances fold sizes. An optional buffer also drops training samples within distance r of any test sample. When b exceeds the range of spatial dependence, the model predicting $\hat{\mathbf{p}}_i$ has seen no sample that shares i's local context, and in particular none that shares a locally clustered labelling error.

Smoothed self-confidence. Let $\mathcal{N}_k(i)$ be the k nearest spatial neighbours of i, excluding i. We define the smoothed probability

$$\bar{\mathbf{p}}_i = w\,\hat{\mathbf{p}}_i + (1-w)\,\frac{1}{k}\sum_{j\in\mathcal{N}_k(i)}\hat{\mathbf{p}}_j, \tag{1}$$

and the suspicion score $\sigma_i = 1-\bar{p}_{i,\tilde{y}_i}$. Neighbour *voting* averages the neighbours' given *labels*, which share any clustered error. Smoothing averages the neighbours' out-of-fold *predictions*, which under spatial CV come from models that never saw the local labels. Smoothing therefore pools independent evidence about the true local class and lowers the variance of the single-sample prediction. In a spatially coherent landscape that variance is the main reason correct labels get flagged. For binary flags we apply the confident-learning rule to $\bar{\mathbf{p}}$: class j gets threshold t_j, the mean of $\bar{p}_{\cdot j}$ over samples labelled j, and sample i is flagged if the most probable class clearing its threshold differs from $\tilde{y}_i$. We use $w = 0.5$ and $k = 10$ throughout and study both in Section 7.

Baselines. (i) *Neighbour vote*: the share of $\mathcal{N}_{10}(i)$ whose given label differs from $\tilde{y}_i$. (ii) *Self-confidence* [12]: $1-\hat{p}_{i,\tilde{y}_i}$, which is smoothed self-confidence with $w = 1$. Each model-based score is computed under random and spatial CV.

Geometry validation. The class of a polygon label is worth questioning only once its geometry is sound. `geolabelqa` also checks GeoJSON annotations against RFC 7946 [3] and common topological defects: unclosed rings, too few vertices, repeated vertices, self-intersection, zero area, slivers (Polsby–Popper compactness $4\pi A/P^2$ below a threshold [16]), wrong winding order, holes outside the shell, out-of-range coordinates, duplicate geometries and missing labels. These checks are not evaluated experimentally here.

4 Benchmark

Landscape and features. A 50×50 km study area is rasterised to 200×200 cells. For each of $K = 6$ classes (water, forest, grassland, cropland, urban, bare) we draw white noise, smooth it with a Gaussian kernel ($\sigma = 6$ cells $= 1.5$ km) and standardise it. Each cell takes the class whose field is highest, which yields irregular homogeneous patches a few kilometres across (Figure 1). We sample 4,000 points uniformly at random: 3,000 for training and auditing, and 1,000 with clean labels for downstream testing. Each point gets six surface-reflectance bands (blue to SWIR2) drawn around a class-specific signature with 30% relative noise. Grassland and cropland, and urban and bare soil, have deliberately similar signatures. Reflectance is multiplied by a smooth illumination field (amplitude 15%, length scale 3 km), which mimics atmospheric and acquisition effects and makes nearby samples similar regardless of class. We add NDVI, NDWI and NDBI. On clean labels a random forest reaches about 80% out-of-fold accuracy, so the task is realistic but not trivial.

Noise regimes. Each class has a look-alike class (water→forest, forest→grassland, grassland↔cropland, urban↔bare). At rate $\rho \in \{0.1, 0.2, 0.3\}$ we corrupt exactly ρn labels:

- **Uniform**: random samples are relabelled to a uniformly random other class.
- **Class-pair**: random samples are relabelled to their look-alike class.
- **Clustered**: discs of radius 3 km centred on random samples are added until ρn samples are covered, and every sample inside is relabelled to its look-alike. This models a regionally systematic annotator or imagery error.

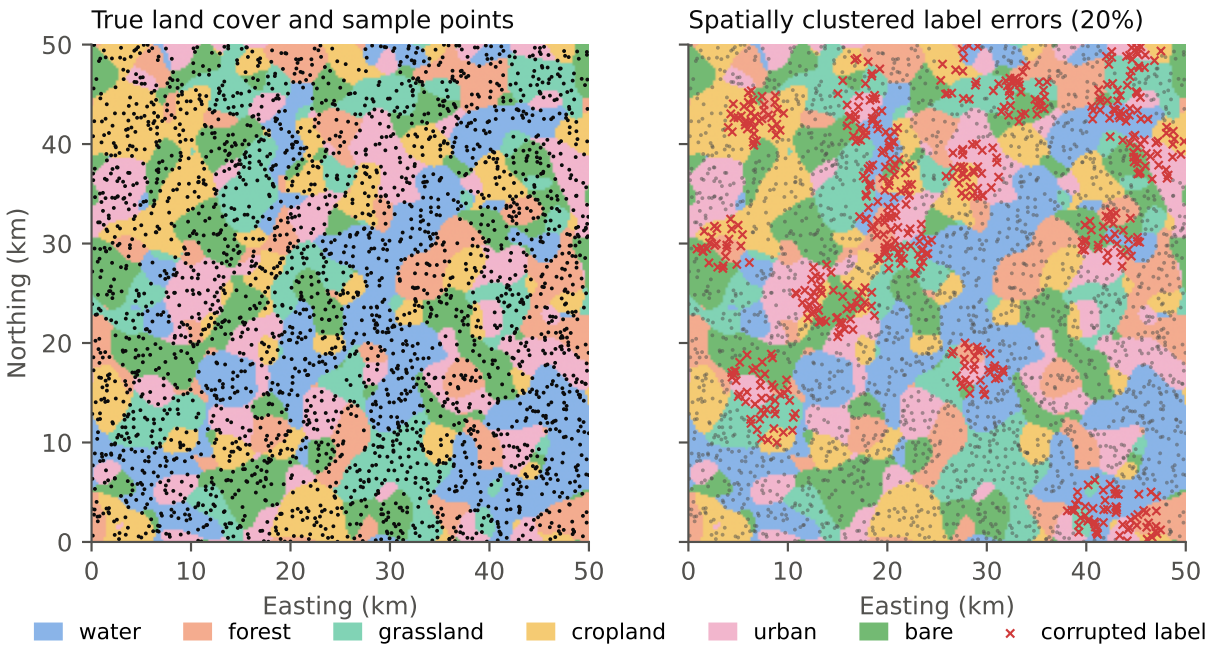


Figure 1: Left: one synthetic landscape with its 3,000 training locations. Right: the same samples after injecting 20% spatially clustered label errors. Each red cross is a sample whose label was changed to its class's look-alike.

Protocol. The classifier is a random forest [1] (100 trees, minimum leaf size 2) from scikit-learn [13], with $F = 5$ folds. Spatial CV uses $b = 10$ km blocks (25 blocks). We compare two feature sets: *spectral* (9 features) and *spectral* + x,y, which appends the coordinates as many geospatial ML pipelines do. Every configuration is repeated with 5 seeds; each seed redraws the landscape, the samples and the noise. We report average precision (AP) for ranking corrupted samples (chance equals ρ), and precision, recall and F1 of the confident-learning flags. For downstream impact we remove the ρn highest-scoring samples (the true error count, so every method gets the same review budget), retrain, and report macro-F1 on the clean test set.

Real scenes. To test whether the findings survive real spectra and real spatial structure, we repeat the protocol on two public hyperspectral scenes with expert ground-truth maps [4]. *Indian Pines* (AVIRIS, 145 × 145 pixels at 20 m, 200 bands after water-absorption removal) is mostly agricultural, with classes laid out as whole fields. We keep the 9 classes with at least 400 labelled pixels (9,234 pixels). *Pavia University* (ROSIS, 610 × 340 pixels at 1.3 m, 103 bands) is urban, with 9 classes (42,776 labelled pixels). The copies we used match the published image sizes and every per-class pixel count. For each seed we draw 3,000 labelled pixels for auditing and 1,000 for testing. Coordinates are pixel centres in metres. The expert ground truth is treated as clean, and noise is injected at a 20% rate. Because the look-alike pairs of real classes are not known in advance, each class's partner is the class that a random forest trained on clean labels most often confuses it with, estimated once per scene on a separate sample. Error discs and CV blocks scale with the scene extent E: disc radius $E/15$ (193 m and 53 m) and block side $E/5$ (580 m and 159 m). Features are the spectral bands, optionally plus coordinates, and downstream models use the spectral bands only. Real label errors in these scenes are unknown, so this is a semi-synthetic test: real features and landscapes, controlled noise.

Table 1: Average precision for ranking corrupted labels at a 20% error rate (mean ± s.d. over 5 seeds; chance = 0.20). Bold marks the best entry per column.

Detector	Features	CV	Uniform	Class-pair	Clustered
Neighbour vote	–	–	0.813 ± 0.014	0.613 ± 0.003	0.234 ± 0.021
Self-confidence	spectral	random	0.908 ± 0.011	0.535 ± 0.008	0.513 ± 0.038
Self-confidence	spectral	spatial	0.902 ± 0.014	0.519 ± 0.021	0.543 ± 0.061
Smoothed SC (ours)	spectral	random	0.944 ± 0.011	0.630 ± 0.023	0.614 ± 0.053
Smoothed SC (ours)	spectral	spatial	0.939 ± 0.009	0.610 ± 0.023	**0.653 ± 0.072**
Self-confidence	spectral + x,y	random	0.931 ± 0.009	0.579 ± 0.019	0.427 ± 0.045
Self-confidence	spectral + x,y	spatial	0.910 ± 0.013	0.523 ± 0.019	0.519 ± 0.061
Smoothed SC (ours)	spectral + x,y	random	**0.957 ± 0.008**	**0.672 ± 0.033**	0.478 ± 0.063
Smoothed SC (ours)	spectral + x,y	spatial	0.940 ± 0.007	0.607 ± 0.025	0.588 ± 0.067

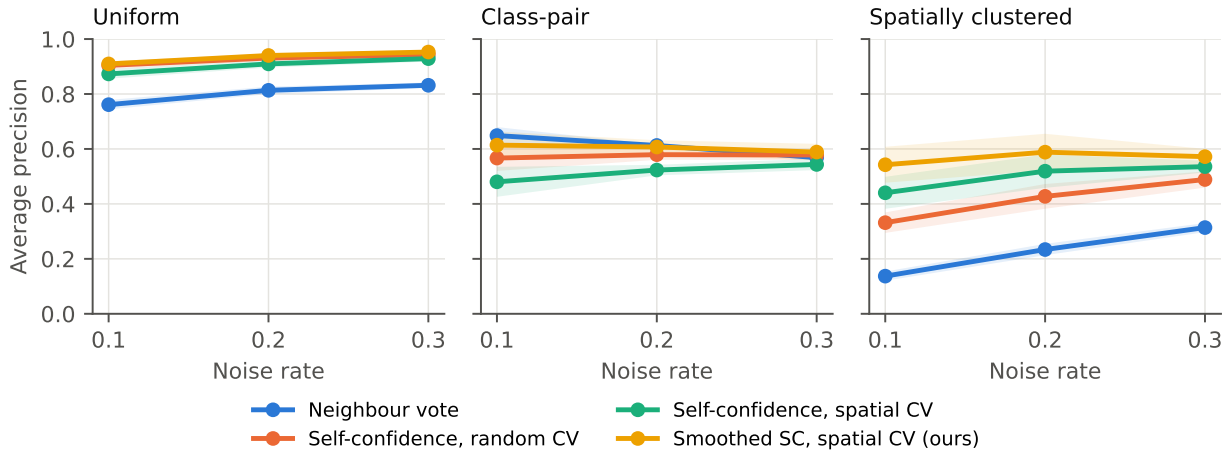


Figure 2: Average precision versus error rate, features = spectral + x,y. Bands show ±1 s.d. over 5 seeds. Clustered errors separate the methods; independent errors do not.

5 Results

Table 1 reports average precision at a 20% error rate for every combination of detector, feature set and CV scheme. Figure 2 shows the trend across error rates for the spectral + x,y setting.

Neighbour voting fails on clustered errors. With uniform errors, neighbour voting is a reasonable detector (AP 0.81). It is also the strongest raw score for class-pair errors at low rates (0.65 at 10%), because an isolated look-alike label stands out in a homogeneous patch. With clustered errors it collapses to AP 0.23 and AUROC 0.57, barely above chance. This is the expected failure, and it is total: a clustered error is surrounded by samples carrying the same error.

Coordinates plus random CV hide clustered errors. Self-confidence with random CV reaches AP 0.51 on clustered errors using spectral features alone. Adding coordinates as features lowers it to 0.43, and to 0.33 at a 10% error rate. The forest learns where the mislabelled patches are and predicts the wrong label back. Spatial block CV removes that shortcut and restores AP to 0.52. With coordinates as features, spatial CV beat random CV in 15 of 15 clustered-noise runs (5 seeds × 3 rates). For spectral features alone the gain is smaller (0.51→0.54), consistent with the

Table 2: Confident-learning flags: precision / recall / F1 at a 20% error rate (spectral + x,y, spatial CV, mean over 5 seeds).

Confident-learning flags on	Uniform	Class-pair	Clustered
Raw probabilities	0.54 / 0.82 / 0.65	0.46 / 0.62 / 0.53	0.41 / 0.54 / 0.46
Smoothed probabilities (ours)	0.61 / 0.80 / 0.69	0.54 / 0.60 / 0.57	0.45 / 0.50 / 0.47

weaker spatial signal that the smooth illumination field carries.

Spatial CV has a cost under independent errors. The same comparison reverses when errors are independent. Random CV beat spatial CV in all 30 uniform and class-pair runs with coordinates, by 0.02 AP (uniform) and 0.06 AP (class-pair) at 20%. With random folds the model sees training samples from the same patch. That helps when those neighbours are correctly labelled, which is the case under independent noise. On this benchmark spatial CV therefore looks like insurance: a small premium when errors are independent, a large payout when they are clustered. Section 6 shows that the premium is much larger on real scenes, where the payout disappears.

Smoothing helps everywhere. Smoothed self-confidence improved AP over raw self-confidence in all 180 runs (3 regimes × 3 rates × 5 seeds × 2 feature sets × 2 CV schemes). With coordinates and spatial CV, the gains at 20% were +0.03 (uniform), +0.08 (class-pair) and +0.07 (clustered). With coordinates and random CV the clustered-noise gain was smaller (+0.05), because the neighbours' predictions share the leaked error. The best clustered-noise result overall is smoothed self-confidence with spectral features and spatial CV (AP 0.65, precision 0.61 at the true error count).

Confident-learning flags. Table 2 applies the confident-learning flag rule to raw and smoothed probabilities (spectral + x,y, spatial CV). Smoothing raises precision in every regime, for example 0.54→0.61 on uniform errors, at a small cost in recall, and F1 rises slightly. The flags recover about half of clustered errors at 0.45 precision, so a human reviewer should treat them as a work queue, not as automatic corrections.

Downstream accuracy. Table 3 reports test macro-F1 after removing the ρn most suspicious samples. The results are a negative finding. The random forest is robust to 20% uniform or class-pair noise (0.805 and 0.789 against 0.812 with clean labels), so there is little to recover. For clustered noise there is a real gap (0.759 against 0.810 when every corrupted sample is removed), but no detector closes it. Smoothed self-confidence gains only 0.004, and neighbour voting makes matters worse. At AP ≈ 0.6, removing the top-n discards many correct samples near the corrupted patches as well as the corrupted ones. The rest of each corrupted patch is left in place, and it still teaches the error. Better detection rankings do not by themselves translate into better models. Relabelling flagged samples, rather than removing them, and auditing iteratively are natural next steps.

6 Results on real scenes

Table 4 repeats the detection comparison on Indian Pines and Pavia University, and Table 5 gives paired Wilcoxon signed-rank tests for the key comparisons on both benchmarks.

Table 3: Test macro-F1 on clean labels after removing the top-n suspicious training samples (n = true error count), 20% error rate, spectral + x,y features (mean ± s.d. over 5 seeds).

Training labels	Uniform	Class-pair	Clustered
No cleaning	0.805 ± 0.025	0.789 ± 0.022	0.759 ± 0.027
Remove top-n: neighbour vote	0.815 ± 0.020	0.807 ± 0.024	0.740 ± 0.027
Remove top-n: SC, random CV	0.807 ± 0.016	0.781 ± 0.011	0.758 ± 0.015
Remove top-n: SC, spatial CV	0.798 ± 0.020	0.778 ± 0.012	0.759 ± 0.013
Remove top-n: smoothed SC, spatial CV (ours)	0.804 ± 0.018	0.779 ± 0.016	0.763 ± 0.013
Oracle: remove all corrupted	0.810 ± 0.019	0.810 ± 0.019	0.810 ± 0.020
Oracle: all labels correct	0.812 ± 0.015	0.812 ± 0.015	0.812 ± 0.015

What replicates. Neighbour voting behaves as on synthetic data, but more extremely. The real scenes are labelled as homogeneous fields and the samples are dense, so isolated errors are almost perfectly detected (AP 0.94–1.00 for uniform noise). Clustered errors, however, fall to AP 0.28 (Indian Pines) and 0.22 (Pavia), with AUROC 0.62 and 0.53. Smoothing also replicates. It improved AP in 107 of 120 paired comparisons, and with spectral features and random CV the gains were +0.03, +0.10 and +0.03 for uniform, class-pair and clustered noise, all significant ($p \leq 0.027$).

What does not replicate. Spatial block CV is worse than random CV on both real scenes in every regime. The losses are large and significant for independent errors (−0.31 and −0.37 AP, 0 of 10 wins), and there is no significant difference for clustered errors (−0.06, 3 of 10 wins, $p = 0.16$). Adding coordinates under random CV did not hurt clustered-error detection on the real scenes either. It gave a small gain (Indian Pines 0.27 → 0.29, Pavia 0.37 → 0.38), unlike the drop on synthetic data.

Why: block transferability. Table 6 reports the out-of-fold accuracy of the same classifier on *clean* labels under each fold design. On the synthetic landscape a held-out 10 km block is almost as predictable as a random sample (a 1-point drop), because each class has one spectral signature everywhere. On the real scenes accuracy drops by 24 points. Fields of the same crop, or roofs of the same material, differ spectrally from place to place, and the blocks remove whole fields. A model that cannot predict a held-out block's correct labels cannot single out its wrong ones, so the out-of-fold probabilities lose most of their discriminative value. The leakage that spatial CV removes is real, but on these scenes it costs less than the transfer that spatial CV gives up. The gap is cheap to measure before an audit, using the given labels, and we propose it as a diagnostic. When the gap is small, spatially blocked folds are the safer choice. When it is large, random folds with spatially smoothed scores are better.

Clustered errors remain hard. The best clustered-error AP was 0.31 on Indian Pines (smoothed self-confidence, random CV, with coordinates) and 0.41 on Pavia (smoothed, random CV, spectral only), against a chance level of 0.20. No method we tested detects regionally systematic errors reliably on real spectra.

Downstream accuracy. Table 7 reports test macro-F1. Removing the top-n suspicious samples *lowered* accuracy under clustered noise for every detector. For example, on Indian Pines removal moved it from 0.719 to between 0.604 and 0.710, against 0.833 for removing exactly the corrupted

Table 4: Average precision for ranking corrupted labels on the real scenes at a 20% injected error rate (mean ± s.d. over 5 seeds; chance = 0.20). Bold marks the best entry per column and scene.

Detector	Uniform	Class-pair	Clustered
Indian Pines			
Neighbour vote	**0.998 ± 0.001**	**0.970 ± 0.007**	0.279 ± 0.025
SC, random CV	0.940 ± 0.006	0.484 ± 0.018	0.268 ± 0.035
SC, spatial CV	0.636 ± 0.030	0.247 ± 0.010	0.225 ± 0.054
Smoothed SC, random CV	0.981 ± 0.001	0.572 ± 0.020	0.283 ± 0.054
Smoothed SC, spatial CV	0.748 ± 0.063	0.255 ± 0.013	0.239 ± 0.077
SC, random CV, +x,y	0.971 ± 0.005	0.623 ± 0.019	0.288 ± 0.041
SC, spatial CV, +x,y	0.713 ± 0.053	0.266 ± 0.012	0.219 ± 0.046
Smoothed SC, random CV, +x,y	0.993 ± 0.002	0.762 ± 0.025	**0.313 ± 0.067**
Smoothed SC, spatial CV, +x,y	0.793 ± 0.071	0.277 ± 0.013	0.227 ± 0.058
Pavia University			
Neighbour vote	0.944 ± 0.008	0.780 ± 0.011	0.222 ± 0.031
SC, random CV	0.963 ± 0.005	0.685 ± 0.007	0.370 ± 0.090
SC, spatial CV	0.595 ± 0.069	0.360 ± 0.036	0.338 ± 0.073
Smoothed SC, random CV	0.988 ± 0.002	0.805 ± 0.019	**0.412 ± 0.121**
Smoothed SC, spatial CV	0.613 ± 0.091	0.380 ± 0.042	0.352 ± 0.078
SC, random CV, +x,y	0.985 ± 0.004	0.808 ± 0.009	0.382 ± 0.077
SC, spatial CV, +x,y	0.605 ± 0.075	0.386 ± 0.037	0.374 ± 0.115
Smoothed SC, random CV, +x,y	**0.997 ± 0.001**	**0.898 ± 0.015**	0.400 ± 0.103
Smoothed SC, spatial CV, +x,y	0.606 ± 0.088	0.406 ± 0.041	0.384 ± 0.123

samples. For class-pair noise on Indian Pines, removal by neighbour vote recovered almost all of the gap (0.816 → 0.848, against 0.851 for the oracle). Automatic removal helps only when detection is precise, which on these scenes means isolated errors.

7 Ablations

Smoothing strength. Figure 3 varies the self-weight w and neighbourhood size k. Every regime peaks at $w = 0.5$, and small neighbourhoods ($k = 5$ or 10) are best. At $w = 0$ the sample's own prediction is discarded and AP falls well below the unsmoothed score. The method is therefore a variance-reduction step around the sample's own prediction, not a replacement for it. With $k = 40$ the neighbourhood spans patch boundaries and the gain shrinks. A natural default is k set so that neighbourhoods stay within typical patch sizes.

Block size. Table 8 varies the spatial CV block size for clustered noise. Blocks of 2.5 or 5 km are similar in size to the 3 km-radius error discs and leave the shortcut partly open. 10 km blocks are best. 20 km blocks (only three columns of blocks across the area, for five folds) start to starve the model of training diversity. This mirrors the guidance from spatial model evaluation: blocks should exceed the range of dependence, here the size of the error clusters, but not by so much that folds stop being representative [17, 23].

Table 5: Paired comparisons of average precision at a 20% error rate. Each cell gives the median difference $a - b$, the number of paired runs where a wins, and the two-sided Wilcoxon signed-rank p-value. Synthetic pairs are 3 rates × 5 seeds; real pairs are 2 scenes × 5 seeds.

Comparison (a vs b)	Uniform	Class-pair	Clustered
Synthetic			
Spatial vs random CV (SC, +x,y)	-0.022 (0/15, <0.001)	-0.049 (0/15, <0.001)	+0.066 (15/15, <0.001)
Smoothed vs raw SC (spatial CV, +x,y)	+0.029 (15/15, <0.001)	+0.085 (15/15, <0.001)	+0.070 (15/15, <0.001)
Smoothed vs raw SC (spatial CV, spectral)	+0.038 (15/15, <0.001)	+0.084 (15/15, <0.001)	+0.106 (15/15, <0.001)
Smoothed vs raw SC (random CV, spectral)	+0.037 (15/15, <0.001)	+0.099 (15/15, <0.001)	+0.098 (15/15, <0.001)
Smoothed SC + spatial CV vs standard CL	+0.031 (15/15, <0.001)	+0.079 (15/15, <0.001)	+0.128 (15/15, <0.001)
Real scenes			
Spatial vs random CV (SC, +x,y)	-0.306 (0/10, 0.002)	-0.374 (0/10, 0.002)	-0.060 (3/10, 0.160)
Smoothed vs raw SC (spatial CV, +x,y)	+0.041 (7/10, 0.049)	+0.014 (10/10, 0.002)	+0.010 (7/10, 0.049)
Smoothed vs raw SC (spatial CV, spectral)	+0.056 (9/10, 0.006)	+0.012 (10/10, 0.002)	+0.010 (8/10, 0.014)
Smoothed vs raw SC (random CV, spectral)	+0.034 (10/10, 0.002)	+0.095 (10/10, 0.002)	+0.031 (8/10, 0.027)
Smoothed SC + spatial CV vs standard CL	-0.249 (0/10, 0.002)	-0.236 (0/10, 0.002)	-0.051 (3/10, 0.322)

Table 6: Out-of-fold accuracy of the random forest on clean labels under random and spatial block CV (3,000 samples, mean ± s.d. over 3 seeds).

Dataset	Random CV	Spatial CV	Gap
Synthetic (10 km blocks)	0.784 ± 0.024	0.774 ± 0.021	0.010
Indian Pines (580 m blocks)	0.838 ± 0.002	0.599 ± 0.038	0.239
Pavia University (159 m blocks)	0.874 ± 0.002	0.636 ± 0.018	0.238

8 Discussion and limitations

Practical recommendations. (1) Use neighbour voting as one signal, never alone. It is excellent for isolated errors in homogeneous, densely sampled data, and blind to clustered errors. (2) Smooth out-of-fold probabilities over a small spatial neighbourhood ($k \approx 10$, $w = 0.5$) before ranking. This was the one change that helped on every dataset. (3) Choose the fold design by measuring the random-versus-spatial accuracy gap. Use spatial blocks only if held-out blocks remain predictable; otherwise use random folds. The `geolabelqa` auditor therefore defaults to random folds, with spatial blocking as an option. (4) Use the ranking to direct human review, not automatic removal. When flagged samples cluster in space, reviewing the whole surrounding region is a natural response to clustered errors, although we have not evaluated it.

Limitations. The noise is injected, not observed: even on the real scenes, we treat expert ground truth as clean and corrupt it with simulated processes. Disc-shaped error clusters and a single look-alike partner per class are simplifications; real annotation errors may follow tile boundaries, image dates or class-specific difficulty. The hyperspectral scenes are small and densely labelled, unlike sparse field-survey samples over large regions, and the synthetic landscape turned out to be far more spatially transferable than either real scene. That is why the spatial-CV conclusion reversed, and it shows that synthetic benchmarks for this problem must be calibrated against real transferability. We used one classifier family (random forest) and one sample size, and fixed the block size and k per scene rather than tuning them. The proposed transferability diagnostic is supported here by three datasets and is not yet a calibrated decision rule. It should be tested on more scenes, including datasets with documented real label errors.

Table 7: Test macro-F1 on clean labels after removing the top-n suspicious training samples on the real scenes (n = true error count, 20% error rate, spectral features, mean ± s.d. over 5 seeds).

Training labels	Uniform	Class-pair	Clustered
Indian Pines			
No cleaning	0.851 ± 0.008	0.816 ± 0.006	0.719 ± 0.024
Remove top-n: neighbour vote	0.848 ± 0.003	0.848 ± 0.008	0.710 ± 0.033
Remove top-n: SC, random CV	0.837 ± 0.009	0.773 ± 0.007	0.665 ± 0.026
Remove top-n: smoothed SC, spatial CV	0.808 ± 0.018	0.683 ± 0.020	0.604 ± 0.075
Oracle: remove all corrupted	0.851 ± 0.008	0.851 ± 0.008	0.833 ± 0.016
Oracle: all labels correct	0.858 ± 0.007	0.858 ± 0.007	0.858 ± 0.007
Pavia University			
No cleaning	0.849 ± 0.018	0.821 ± 0.014	0.770 ± 0.027
Remove top-n: neighbour vote	0.846 ± 0.019	0.823 ± 0.030	0.688 ± 0.077
Remove top-n: SC, random CV	0.837 ± 0.014	0.810 ± 0.015	0.735 ± 0.025
Remove top-n: smoothed SC, spatial CV	0.772 ± 0.019	0.736 ± 0.026	0.715 ± 0.024
Oracle: remove all corrupted	0.851 ± 0.014	0.851 ± 0.014	0.830 ± 0.024
Oracle: all labels correct	0.857 ± 0.014	0.857 ± 0.014	0.857 ± 0.014

Table 8: Effect of block size on clustered-error detection (AP, 20% error rate, spectral + x,y, mean ± s.d. over 5 seeds).

Block size (km)	Self-confidence	Smoothed SC
2.5	0.456 ± 0.034	0.513 ± 0.052
5	0.472 ± 0.054	0.529 ± 0.063
10	0.519 ± 0.061	0.588 ± 0.067
20	0.497 ± 0.039	0.569 ± 0.064

9 Conclusion

Spatially clustered label errors defeat neighbour voting and remain hard for model-based detectors. Smoothing out-of-fold class probabilities over spatial neighbours is a cheap, consistent improvement on synthetic and real data alike. Spatial block cross-validation, the standard remedy for spatial leakage in model evaluation, helps label auditing only when held-out regions remain predictable. On two real hyperspectral scenes they did not, and random folds were better. We offer the random-versus-spatial accuracy gap as a simple way to tell the two situations apart. Everything is implemented in `geolabelqa`, which also checks annotation geometry.

Reproducibility. `python experiments/run_benchmark.py` regenerates the synthetic results (about 30 minutes on a laptop CPU). After downloading the two scenes, `python experiments/run_real.py` and `python experiments/transferability.py` regenerate the real-scene results. `python experiments/make_fi` regenerates every figure and table in this paper.

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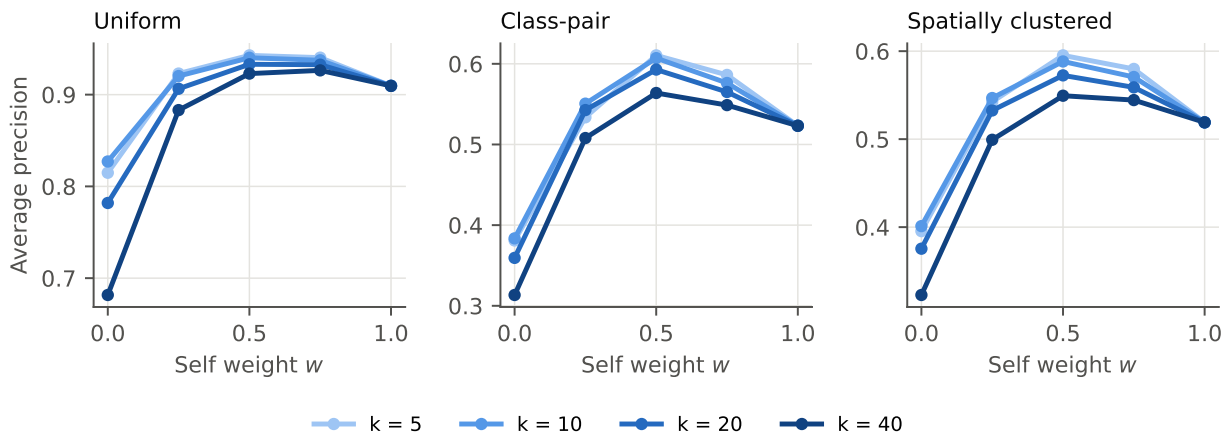


Figure 3: AP of smoothed self-confidence versus self-weight w ($w = 1$: no smoothing) and neighbourhood size k. 20% error rate, spectral + x,y, spatial CV, mean over 5 seeds.

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